

Bas van Fraassen

Arguments Concerning Scientific Realism

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Bas C. van Fraassen

- Born 1941 in Goes
- Princeton
- Constructive empiricism
- “Rehabilitating anti-realism” [1]

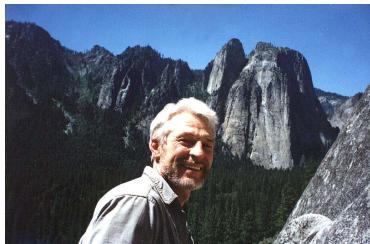


Figure: Bas C. van Fraassen [2]

Scientific Realism

Definition (by van Fraassen)

Science aims to give us, in its theories, a literally true story of what the world is like; and acceptance of a scientific theory involves the belief that it is true [3].

- *Literally*
 - Capable of being true/false
 - Cannot change logical relationships
- Acceptance as belief

Constructive empiricism

Definition (by van Fraassen)

Science aims to give us theories which are empirically adequate; and acceptance of a theory involves as belief only that it is empirically adequate [3].

- Empirically adequate
 - True about observable things and events
- Acceptance
 - Epistemically: agnost beyond the observable
 - Pragmatically: use conceptual terms for explanation

The (Un)observable

Responding to Grover Maxwell

- Distinction theoretical and non-theoretical? → No
- Distinction observable and unobservable? → Yes
 - Observability continuum
 - Observability via limitations as human beings
 - Empirical adequacy as a function of the epistemic community



Figure: Microscope

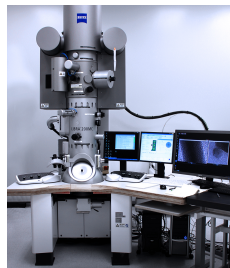


Figure: Electron Microscope

The **No Miracle Argument** (NMA)

Scientific Realism

Our best theories are extraordinarily successful, therefore they are (approximately) true [4]. Otherwise, the success of science would be miraculous.

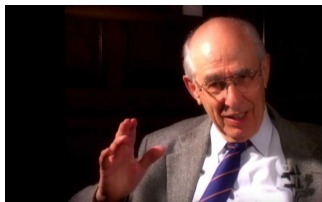


Figure: Hilary Putnam: *If the scientific theories are false, why would they be so successful [1]?*

Bas van Fraassen's Response

For any scientific theory is born into a life of fierce competition, a jungle red in tooth and claw. Only the successful theories survive—the ones which *in fact* latched on to actual regularities in nature [3].

Example (The Mouse who Runs from the Cat)

Do not ask why the mouse runs from its enemy. Species which did not cope with their natural enemies no longer exist. That is why there are only ones who do [5].

Inference to the Best Explanation (IBE)

Scientific Realism

We should acknowledge the reality of the entities our best explanatory theories postulate, even if those entities are unobservable [1].

Bas van Fraassen's Response

IBE is a rule that selects the best among the historically given hypotheses. We can watch no contest of the theories we have so painfully struggled to formulate, with those no one has proposed. So our selection may well be the best of a bad lot [6].

Observable/Unobservable Distinction

Churchland

The beings who are like us except that they are born with electron microscopes permanently attached to their left eyes [1].

Ian Hacking

A machine which makes grids of the same shape but various sizes. We can see grids with the same overall shape of smaller and smaller size, but the machine makes some grids that are too small to be seen with the unaided eye. When looked at through a microscope, however, the unobservable grids are seen to have the same shape as the observable ones [1].

Alspector-Kelly

When we look directly at the tree we are also postulating an appropriate relation between object, image, and vantage point, namely, between the tree itself, our perceptual experience of the tree, and the vantage point of our bodily location [1].

How would Bas van Fraassen respond to these examples? Or, to put it another way, can Constructive Empiricism provide answers to these examples that would satisfy philosophers?

Open-Ended Discussion Questions

- The range of observability. Given the rapid advancements in technology, to what extent does constructive empiricism need to reconsider its stance on observability?
- The debate between scientific realists and constructive empiricists. How does it assist us in understanding the *nature* of scientific knowledge and how we perceive the essence of reality?
- Integration with the Philosophy of Mind. Can discussions about consciousness, perception, and cognition help us further understand the essence of sensation?
Further reading: *Phenomenology of Perception* by Maurice Merleau-Ponty.

Reference

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- [6] Samir Okasha. “Van Fraassen’s critique of inference to the best explanation”. In: **Studies in History and Philosophy of Science Part A** 31.4 (2000), pp. 691–710.
- [7] Maurício Moreira Neto. **Template Beamer UFC**. 2020. URL: <https://www.overleaf.com/latex/templates/template-beamer-ufc/rvqwnmszpsvf>.